

Analog Buck-Boost (LTC3440, 3.9 V): Working Document

Digital Audio Player • Rev 8

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

Contents

1. Summary	1
2. Design Requirements	1
3. Pin Configuration	2
4. External Components	2
5. Equations and Results	2
6. Important Values	3
7. Open Problems	3
8. Revision Log	3
9. Sources	3

1. Summary

Field	Value
Manufacturer / MPN	Analog Devices, LTC3440EMS#TRPBF (#1-050), U200
Package	10-lead MSOP (MS)
Function	Synchronous buck-boost, SYS → 3.9 V intermediate analog rail

2. Design Requirements

Requirement	Value
Input	SYS, 3.0 to 4.2 V
Output	3.9 V
Load	Analog chain only: two LT3042 (+3V3_ANALOG, +3V3_ADC) and LT3462A → LT3093
Output current	≈ 263 mA peak: U204 104 + DAC LDO 10 (8 load + ≈ 2 quiescent) + U204 quiescent ≈ 2 + LT3462A input 147; 250 mA design current exceeded by ≈ 5 %
Fault ceiling	≈ 0.31 A (U204 at 125 mA + DAC LDO + LT3462A at its switch limit)
Switching frequency	1.2 MHz, fixed (Burst Mode off)
Output ripple target	0.5 % (19.5 mV pp)
Soft-start	Output up in ≈ 6 ms, MCU-enabled

Requirement	Value
Constraints	Low noise · Small footprint

Source: [3]

3. Pin Configuration

Pin	Name	Connection
1	RT	R_T to GND
2	MODE/SYNC	GND (fixed frequency)
3, 4	SW1, SW2	L
5	GND	GND
6	VOUT	+3V9, C_OUT, R1
7	VIN	SYS, C_IN
8	SHDN/SS	R_SS from GPIO35 (1 MΩ pull-down on the GPIO side), C_SS to GND
9	FB	R1 / R2 midpoint
10	VC	C_P1 and R_Z to FB

4. External Components

Ref	Value / Rating	MPN	AL-PN
L (L200)	6.8 μH ±20 %, I _{sat} 3.5 A, 74.1 mΩ	Bourns SRP4030FA-6R8M	#1-076
C_IN (C203)	10 μF 10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
C_OUT (C204)	22 μF 25 V X7S 1210	Murata GCM32EC71E226KE36L	#1-073
R1 (R200)	220 kΩ 0.1 % 0402	YAGEO RC0402BR-07220KL	#1-074
R2 (R212)	100 kΩ 0.1 % 0402	YAGEO RT0402BRD07100KL	#1-131
R_T (R207)	50 kΩ 0.1 % 0603	YAGEO RT0603BRD0750KL	#1-075
R_Z (R213)	15 kΩ 0.1 % 0402	YAGEO RT0402BRD0715KL	#1-078
C_P1 (C212)	1.5 nF 2 % C0G 0402	Murata GRM1555C1H152GA01D	#1-077
R_SS (R203)	54.9 kΩ 0.1 % 0402	YAGEO RT0402BRD0754K9L	#1-079
C_SS (C213)	0.1 μF 50 V X7R 0402	Murata GCM155R71H104KE02D	#1-080
R_PD (R308, MCU sheet)	1 MΩ 0.1 % 0402	YAGEO RE0402BRE071ML	#1-123

Source: [2]

5. Equations and Results

$$R_T = 6 \times 10^{10} / 1.2 \text{ MHz} = 50 \text{ k}\Omega; \quad V_{OUT} = 1.22 (1 + 220/100) = 3.904 \text{ V} \quad (1)$$

$$Ripple_{boost} = \frac{3.0 \times 0.9}{1.2 \times 0.25 \times 6.8 \times 3.9} = 0.34 (85 \text{ mA}_{pp}); \quad Ripple_{buck} = 0.14 \quad (2)$$

$$I_{L, pk} = \frac{0.263 \times 3.9}{3.0 \times 0.85} + \frac{0.085}{2} = 0.44 \text{ A} \quad (3)$$

$$\tau = R_{SS}C_{SS} = 5.49 \text{ ms} : \text{enable } (1.0 \text{ V}) \text{ } 2.0 \text{ ms}, \text{ ramp complete } (2.2 \text{ V}) \text{ } 6.0 \text{ ms} \quad (4)$$

$$V_{IN} : V_{OUT}(1 - 150 \text{ ns} \cdot f) \text{ to } V_{OUT}/(1 - 150 \text{ ns} \cdot f) = 3.20 \text{ to } 4.76 \text{ V} \quad (5)$$

Source: [1]

Case	U204	DAC LDO	LT3462A input	LTC3440 load
Quiescent	≈ 34 mA	≈ 10 mA	≈ 38 mA	≈ 82 mA
Full scale, 23.8 Ω	≈ 106 mA	≈ 10 mA	≈ 147 mA	≈ 263 mA
Fault	125 mA	≈ 10 mA	≈ 0.18 A	≈ 0.31 A

U204 figures include its ≈ 2 mA ground current. Previous revisions omitted the DAC oscillators (≤ 5.5 mA) from the +3V3_ANALOG load.

6. Important Values

Item	Value
Switch current limit	2.7 A typ (2.5 to 2.9 A)
Simulated output ripple	7.7 mV pp (target 19.5 mV)
Load step 150 ↔ 250 mA	±90 mV, no ringing, 0.3 ms recovery
Minimum rail during step	3.74 V
Fault ceiling	≈ 0.31 A: inductor peak ≈ 0.5 A vs 3.5 A I _{sat}

7. Open Problems

#	Problem
1	Load is ≈ 5 % above the 250 mA design current; ripple and step response were simulated at 250 mA. Re-check the load step at 263 mA or accept the margin.

8. Revision Log

Rev	Date	Change
0 to 4	2026-09-21 to 23	Initial design, simulation, load updates, inverter branch.
5 to 7	2026-09-25 to 27	Reformatted; LT3462A efficiency; load from Power Budget Rev 6 (253 mA).
8	2026-10-03	Second LT3042 (DAC rail) and oscillator load added: ≈ 263 mA peak. C _{IN} 10 μF (#1-081); R2 0402 (#1-131). Designators added. Added: load margin item.

9. Sources

1. LTC3440 Product Page and Datasheet (Analog Devices)
2. DAP schematic sheet DAP-Power-2, 3.9 V buck-boost (author, 2026-10-03)
3. Power Budget, Rev 7